

Interaction of epoxy-based hydrogels and water: A molecular dynamics simulation study

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ABSTRACT

Biomaterials play a crucial role in tissue engineering as a functional replacement, regenerative medicines, supportive scaffold for guided tissue growth, and drug delivery devices. The term biomaterial refers to metals, ceramics, and polymers account for the vast majority. In the case of polymers, hydrogels have emerged as active materials for an immense variety of applications. Epoxy-based hydrogels possess a unique network structure that enables very high levels of hydrophilicity and biocompatibility. Hydrogel such as Medipacs Epoxy Polymers (MEPs) models were constructed to understand water's behavior at the water/hydrogel interface and hydrogel network. We computed the Gibbs dividing surface (GDS) to define the MEP/water interface, and all the physicochemical properties were computed based on GDS. We calculated the radial distribution function (RDF), the 2D surface roughness of the immersed MEPs. RDF analysis confirmed that the first hydration shell is at a distance of 1.86 Å, and most of the water molecules are near the hydroxyl group of the MEPs network. Hydrogen bonds (H-bonds) analysis was performed, and the observation suggested that the disruption of the H-bonds between MEP chains leads to an increase in the polymer matrix's void spaces. These void spaces are filled with diffused water molecules, leading to swelling of the MEP hydrogel. The swelling parameter was estimated from the fitted curve of the yz-lattice of the simulation cell. The MEP/water interface simulation results provide insightful information regarding the design strategy of epoxy-based hydrogel and other hydrogels vital for biomedical applications.

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1. Introduction

Biomaterials like hydrogels have been recognized as essential biomaterials in the field of tissue engineering, regenerative medicine, and drug delivery applications due to their specific characteristics [1,2]. Hydrogels represent the cross-linked polymer matrices capable of holding a large amount of water in their three-dimensional network and swollen up to thermodynamic equilibrium [3,4]. Since the 1960s, hydrogels have been widely investigated in research labs and used in industry as essential biomaterials due to their outlandish combination of attractive properties [3,5–7]. Hydrogel properties can be tailored by appropriately changing selected characteristics, e.g., polymer composition or cross-link density, leading to their wide range of applications from

tissue engineering [8] to devices for controlled drug release [9–11]. Besides, hydrogels can be designed as smart materials that can respond to various environmental stimuli such as temperature, pH, electric fields, light, and enzymes [12,13]. For instance, epoxy-based electroactive polymer hydrogel has been studied and developed a pulse actuated cell system architectures in drug delivery as infusion pumps [14].

Epoxy-based polymeric materials, synthesized from the reaction between resins and cross-linking agents such as amines, have become a focal component in many industries and engineering applications due to its various properties [3,15,16]. Among different types of amines, Jeffamines with three primary amines is showing unique characteristics [17]. Many experimental and computational studies have been reported in the literature [18] [–] [28]. However, most of the work addressed the effect of moisture on mechanical properties using different percentages of water in the hydrogel matrix [18,19,21–27]. A spectroscopic study by Mijovic et al. [26] confirmed that 95% of water molecules reside in the vicinity of polar groups present in epoxies, and equilibrium water content is

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governed by the number of available polar groups in the matrix. Salahshoor et al. [23] employed the coarse-grained based molecular dynamics (MD) simulations to study the effect of water on the equilibrated structure and the mechanical properties of the cross-linked hydrogel at multiple scales. However, in this study, the structure of hydrogel was directly equilibrated for various volume percent of water content, regardless of water absorbency. Agrawal et al. [24] described a new class of hydrogel composites reinforced with elastic fibers to form a cartilage-like structure. Their experiments determined the saturated water contents of the reinforced and unreinforced hydrogels but did not describe the hydrogel's water-absorbing mechanism. A few research works have focused on the water absorbency and absorbing mechanism of the hydrogel in a nanoscale. Xiang and co-workers [25] used MD to study water absorption on the surface of a hydrogel prepared from PEGDGE as the epoxy resin and Jeffamine as the curing agent at different cross-linking degrees and comparing the water absorption curves at different temperatures. The simulation outcomes showed that the hydrogel's water absorption content and its surface reached its saturation point-both temperature and degree of cross-linking influence the saturation water content of the hydrogel and its surface. Most of the MD studies of hydrogels network were done by equilibrated hydrogel structure with specific volume or weight percent of water content, ignoring the saturated water contents of the hydrogel.

In this work, we model hydrogel, which consists of Polyethylene glycol diglycidyl ether (PEGDGE) as epoxy and the Jeffamine T-403, poly-oxy-alkylene- amines, as a curing agent. All atom-based molecular dynamics simulations are used to understand water absorption, diffusion mechanism and to estimate the swelling of hydrogel under room temperature and pressure. We modeled four different hydrogels/water interfaces based on the curing agents' structural constraints as per the experimental evidence.

2. Modelling and computational methods

The system studied consisted of poly ethylene glycol diglycidyl ether (PEGDGE) unit as the epoxy monomer and JEFFAMINE T-403 (Tri-amine) as a cross-linking agent with the ratio 1.25:1 in which amine hydrogens to the epoxy group is related to curing reaction [14,29]. The chemical structure of these molecules is depicting in Fig. 1. The cross-linked epoxy polymer or Medipacs epoxy polymer (MEP) containing the EGDGE with $n = 1$ is shown and it is subject to the structural restrictions: $x + y + z \approx 5.3$. There are four different structures possible for the tri-amine, based on the oxypropylene units $[-OCH_2CH(CH_3)-]$ in the amine chain containing the cross-linked amine groups, JEFFAMINE T-403. In Fig. 2, we display all the possible six configurations of MEP based on the constraint of cross-linker but we did not include the MEP (122) = MEP (212) model in the discussions as the potential energy never converged. The cross-link density, ρ , of each cross-linked chain is 0.66 and due to the steric hindrance, secondary amine hydrogens will be predominant in polymers [26].

To understand MEPs' behavior when hydrated with water or electrolytes, we performed classical atomic-scale molecular dynamics (MD) simulations under a dry conditions and for MEP embedded in a water interface. We have described all the details of three MEP/water models, as we found that in all our trials MEP (122) is highly unstable and exploded after several attempts. It might be related to unfavorable dipole arrangements, nonetheless we are aware that a favorable configuration of MEP (122) could exist but its divulge was beyond applied here computational methodology. The initial dry MEP structures were constructed with high crystallinity where six chains of MEPs were fully stretched and aligned along the x -axis. Subsequently they were arranged in a

hexagonal pattern across the yz plane to get a bulk structure with high density; two slabs were juxtaposed along the x -axis in the middle of the simulation box with varied density from 1.05 to 1.10 g/cm³. The resulting system consists of 12 chains of MEP totaling 2976 atoms. To saturate the epoxy polymer a different number of TIP3P water molecules were used. The description of the number of atoms and size of the simulation box are depicted in Table 1. The shape of the simulation box is described in Figure S1.

CHARMM27 force field parameters were used for MEPs [30,31]. The force field parameter of MEPs were generated by using automated topology builder SwissParam [32] and water was modeled using the TIP3P model [33]. Although SwissParam is mainly designed for small molecule drug-like molecule, it is widely used in polymeric materials [34,35]. The partial charges assigned to all the polar sites that appear in the modeled MEPs and the TIP3P water molecule are displayed in Table 2 and Figure S2.

Classical MD simulations of MEP/water systems were performed using GROMACS-4.6 package [36]. In the initial state we conducted energy minimization of the system that consists of 12 chains of MEPs hexagonally packed (cf. Figure S2) using the steepest descent method. It was followed by dry NVT equilibration for 4 ns at 300 K to achieve uniform distribution of the polymer. In the next steps, we solvated the polymer with the TIP3P water model and removed the water molecules from the bulk polymer region so that the polymer matrix does not contain any water molecules. Finally, we performed energy minimization of the solvated system followed by NVT equilibration for 2–4 ns at 300 K with position restraint on water molecules. Before the final MD run, we performed two sets of NPT equilibrations for another 5–12 ns at 300 K with position restraint on water molecules and heavy atoms of polymers, successively (see Supporting information). The temperature was controlled by the velocity rescale method [37] with a coupling constant of 0.1 ps, while pressure was controlled by the Berendsen method [38] with a coupling constant of 1.0 ps. Short-range van der Waals (VDW) forces and electrostatic forces in real space were switched to 0.7 nm. We used the particle mesh Ewald (PME) method with a Fourier spacing of 0.12 nm for the long-range calculation of electrostatic forces. To accelerate the calculation, we adopted the LINCS algorithm to constrain the bonds. Production MD simulations were conducted from the equilibrated structure for another 180 ns with allowing both the water and polymer to be relaxed with a typical sampling frequency of 1 ps by using NPT ensemble, and with a time-step of 2 fs. The V-rescale thermostat and Parrinello–Rahman barostat [39] were used to control the temperature and pressure with a coupling constant of 0.1 ps and 3.0 ps, respectively. The molecular viewer software package VMD was used for graphic visualization [40].

The 2D surface roughness parameters MEP polymers computed by dividing the yz -plane of the simulation box in small grid elements of 10 Å × 10 Å. The surface topology obtained by taking the topmost atoms x -coordinate in each xy grid element. The more complex profile root mean squared, R_q , method adopted to evaluate 2D surface roughness as defined by Equation (1) [41,42]:

$$R_q = \sqrt{\frac{1}{N} \sum_{i=1}^N \Delta z_i^2} \quad (1)$$

where N is the number of surface atoms and Δz_i their distances from the mathematical surface or Gibbs dividing surface (GDS). The roughness of the polymer in conjunction with the GDS will be used to define the polymer slab thickness.

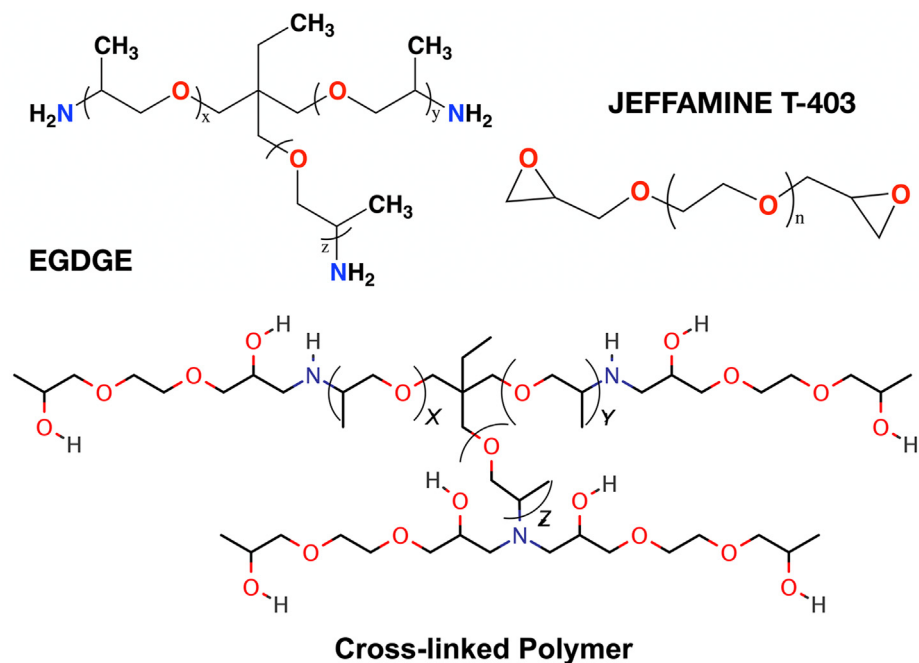


Fig. 1. Chemical structures of the EGDGE (epoxy monomer), JEFFAMINE T-403 (cross-linker) and cross-linked epoxy polymer network or Medipacs epoxy polymer (MEP) where $n = 1$ for EGDGE. In this polymer, $x + y + z \approx 5.3$.

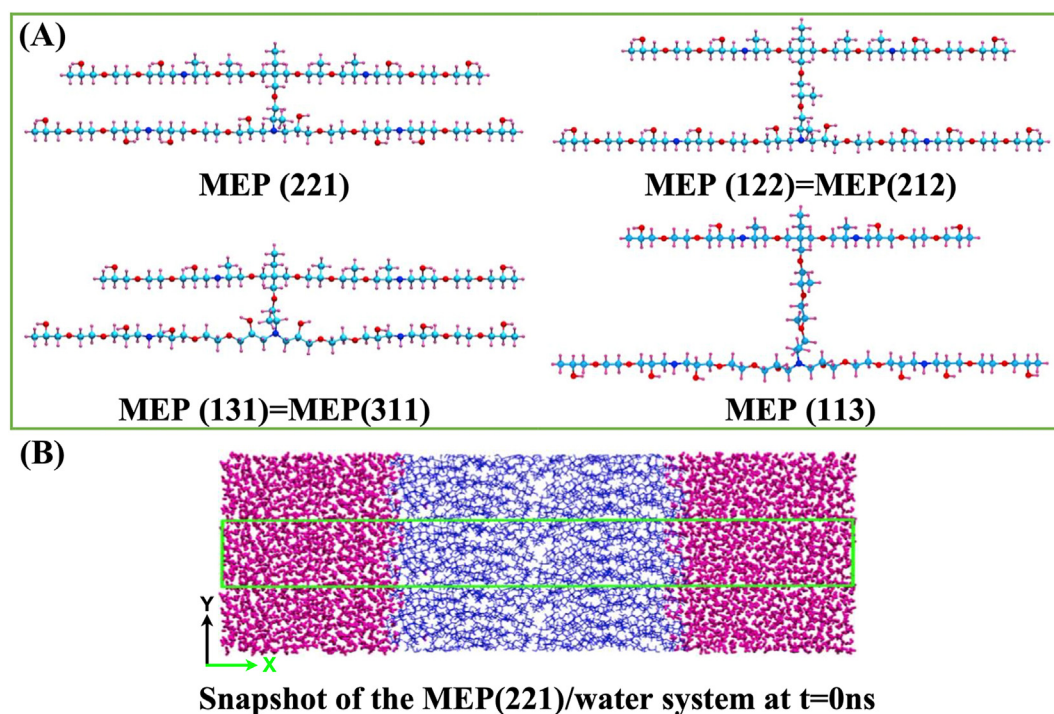


Fig. 2. (A) Atomic structure of all four possible models of MEP Color code: hydrogen, carbon, oxygen and nitrogen atoms are in mauve, cyan, red and blue, respectively. (B) Side view of snapshot of the MEP (221)/water from the MD simulations at $t = 0$ ns with axes (to guide eye), where green box represents the simulation cell. Blue and red regions represent the bulk MEP (221) and bulk water respectively. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

3. Results and discussion

3.1. Structural properties

We begin with the analysis of the polymer and water structure at the interface via analysis of the density profiles for water and

polymer atoms extracted as a function of the x -axis of the simulation box, which is defined as the normal to the polymer-water interface i.e., yz -plane of the simulation box. The initial ($t = 0$ ns) and final ($t = 180$ ns) density profiles of the three systems are depicted in Fig. 3. We used *gmx density* tools to get partial mass density profiles along the x -axis. The simulated average bulk

Table 1

Details of the systems simulated in this study: number of atoms and simulation lattice parameters before energy minimization and molecular dynamics.

System	Number of atoms			Lattice parameters, ($a \times b \times c$; $\alpha = 120, \beta = \gamma = 90$)
	MEP	TIP3P	Total	
MEP (221)/Water	2976	3423	6399	$24 \times 1.72 \times 1.49$ (Å)
MEP (131)/Water		3456	6432	$24 \times 1.69 \times 1.51$ (Å)
MEP (113)/Water		5379	9355	$24 \times 1.54 \times 2.22$ (Å)

Table 2Partial charge in *e* units assigned to the polar sites of epoxy polymer MEPs and TIP3P water molecule.

Polar Sites	MEPs ³²	TIP3P ³³
Tertiary amine (>N-)	-0.81	
Secondary amine (-NH-)	-0.94	
Ether	-0.56	
Oxygen (-OH)	-0.68	-0.834

density of water varied from 0.74 to 0.78 g/cm³ (Table 3, Fig. 3),

which is 22% less than the TIP3P water model density 0.98 g/cm³ [43]. The lower density of water originates due to the i) diffusion of water molecules into the polymer matrix affecting the volume of the simulation box, and ii) the position of water molecules at both sides of the polymer matrix that increases the volume occupied by water in long simulation cell.

The density of the polymer fluctuated from 0.87 to 0.95 g/cm³ (cf. Supporting Information, Figure S4). During the simulations, we observed that the slope of the water density curve near the polymer surface decreased with the evolution of time. Also, the stepwise approach of the water density curve at the interface suggested that

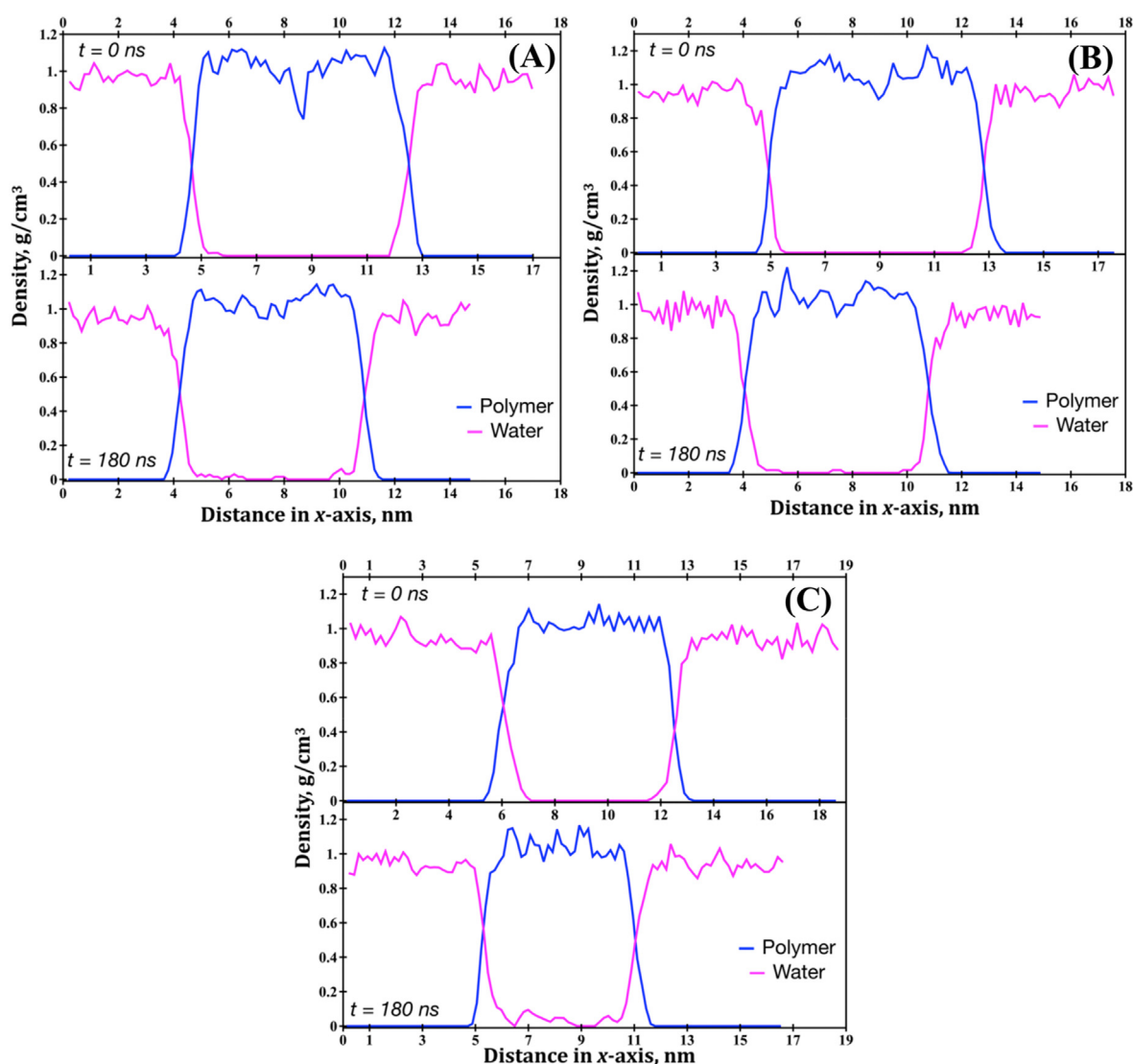


Fig. 3. Partial mass density profile of MEP (221) (A), MEP (131) (B) and MEP (113) (C) for polymer-water interface at 300 K. Density was computed as a function of the x-axis distance of the simulation box.

Table 3

Computed parameters of MEP-water interfaces simulated in this study: average roughness, average GDS position, and average density.

	Average Roughness (nm)		Average GDS Position (nm)		Average Density (g-cm ⁻³)	
	Left	Right	Left	Right	Water	Polymer
MEP (221)/Water	0.938	0.968	4.181	10.986	0.746	0.913
MEP (131)/Water	1.022	0.951	4.163	11.085	0.779	0.927
MEP (113)/Water	0.940	0.813	5.304	11.274	0.735	0.874

water molecules are clustered or somewhat ordered at the interface. This phenomenon is because some parts of the MEP are hydrophilic which might facilitate the formation of hydrogen bonding or other weak interaction forces.

Fig. 3 gets a clear view of the behavior (diffusion and translational ordering) of water molecules and how they cover the distances into the bulk polymer during the simulation time. We observed that in the bulk polymer region, the peak widths of the water density profile describes the cluster size. At the same time, the intensity gives an idea about the number of water molecules. The depth of diffusion of water molecules into bulk polymer reveals information related to the thickness of the water-polymer interface. Also Fig. 3, indicates that the length of simulation box is decreasing with the growing simulation time due to the swelling of the simulation box in yz-direction.

3.2. Water-Polymer Interface

Interfaces of solid and liquid are not well-defined due to the surface roughness of solid. For this reason, it is crucial to define a theoretical surface called Gibbs dividing surface (GDS) [42,44–47]. Vaecha et al. define the interface of hydrophobic droplet-water interface using GDS [47]. Muntean et al. also use the roughness of polystyrene and GDS to describe the thickness of interface between the water and polystyrene [42]. The thickness of the interface can be computed from GDS and surface roughness. The position of GDS is estimated by changing the polymer density profile into a step-like profile while the bulk density and the number of particles remain constant. A graphical illustration of this procedure is given in Fig. 4. We define the interface by the relation, $interface = GDS \pm \frac{1}{2}R_q$. If one uses data from Table 3 and applies this relation, the average interface thickness will be 2 nm.

Computed parameters like average roughness, average GDS position and average density of the total simulation time are tabulated in Table 3. The simulated average roughness is lower in the symmetric model of MEPs such as MEP (221) and MEP (113), compared to the asymmetric MEP (131). The roughness of the simulated polymer surfaces decreases upon simulation time, and after a specific time, it fluctuates to less than 0.1 nm (cf. Supporting Information, Figure S5). This effect might be due to the exposure of more oxygen and nitrogen atoms of the MEPs to the water, reducing the roughness [42]. Trudeau et al. suggested that the orientation of water molecules in contact with hydrophilic surfaces are stable due to the dipole-dipole interactions between water molecules and the polar part of the organic compound [48].

To describe the structure of water molecules in different regions such as bulk water, polymer/water interface, and bulk polymer, we computed the orientational order parameter (S_q) of water molecules in each region by using the following definition:

$$S_q = \frac{3}{2} \cos^2 \theta - \frac{1}{2} \quad (2)$$

where θ is the angle between the water dipole moment and the interface normal (x -axis) as described in Fig. 5. The value of S_q is zero when the orientational distribution is flat (no directional preference), S_q is near unity when the water molecules are strongly oriented perpendicular to the x -axis; S_q near -0.5 indicates that water molecules are parallel to the interface [49].

The computed order parameters of water molecules in different regions of the MEP (221)/water system are depicted in Fig. 5. Order parameters of MEP (131) and MEP (113) are described in supporting information, Figure S6. In all the cases, the values of order parameters in bulk water, as expected, are around zero. However, the orientation of water molecules inside the polymer matrix has a prominent direction i) out of the surface plane, as indicated by the positive values and ii) negative values. The latter case is prevalent in the time range between 0 and 30 ns and throughout the evolutions of time, affirming the interface parallel orientation of the water molecules. This observation establishes the hypothesis that orientation of water molecules improves H-bond formation possibilities when the number of oxygen and nitrogen atoms increases in their vicinity. In the polymer/water interface, water molecules have an inferior orientation to form H-bonds, which is slightly higher than the bulk water because of the higher density of alkyl chains. This order originates from the hydrogen bonding between water molecules and the hydrophilic $-OH$, ether oxygen, $-NH$ and N of the polymer entity.

3.3. Structure of water

We computed the RDF between the water molecules' oxygen atoms and the hydrogen atoms of hydroxyl and amine groups of MEPs. Partial RDF of $O \cdots H$ pairs (H_2O-HN/O) and $O \cdots O$ pairs (H_2O-O , ether) of the MEPs model are depicted in Fig. 6. For $O \cdots H$ pairs of all the MEP models have two prominent peaks centered at 1.86 nm and 3.35 nm, which indicates the two hydration shells structure. The intensity of the RDFs first peak at confirms that the water molecules are bounded by hydrogen bond to all the available sites of the MEPs. The second hydration shell can be inferred from second RDF peak intensity [50,51]. The one prominent RDF peak for the $O \cdots O$ pairs at 2.78 nm indicates the distance between the neighboring water molecules and the backbone oxygens of epoxy network. RDFs revealed not only the formation of H-bonds between the polar hydrogens of MEP and oxygen of water molecules, but also strength of H-bonds [52].

As the degree of cross-link is the same, the number of polar sites is also constant for all the models, resulting in an almost similar RDF. The peak intensity is lowest for MEP (113), which could be due to free volume from its morphology. RDF analysis confirmed that the stability of water molecules facilitated by the vicinity of hydroxyl groups.

3.4. Hydrogen bond dynamics

To understand evolution of the molecular interactions between water molecules and epoxy networks, the formation of H-bond and its behavior has been explored. The following two geometrical criteria were used herein to define a H-bond: i) the distance between hydrogen and an acceptor is less than 1.86 Å and ii) the angle between hydrogen and an acceptor is larger than 35° [19]. The H-bond cutoff distance corresponds to the RDF minimum location, as described in Fig. 6. This interaction is essentially controlled by the hydrogen-bonding sites in the polymer networks [53] and plays a

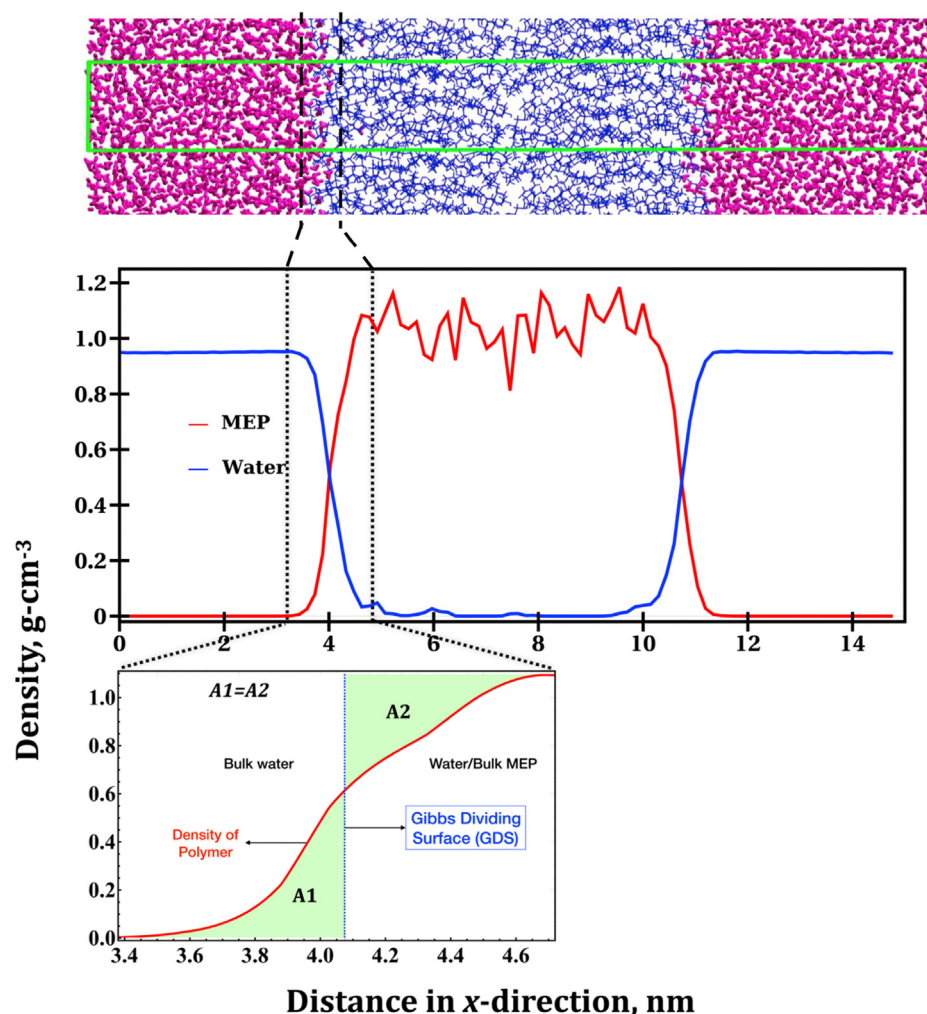


Fig. 4. Graphical definition of Gibbs dividing surface (GDS) based on the density of MEP at the interface as a function of the length of the x-axis of the simulation box. The two green shaded areas are equal in size. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

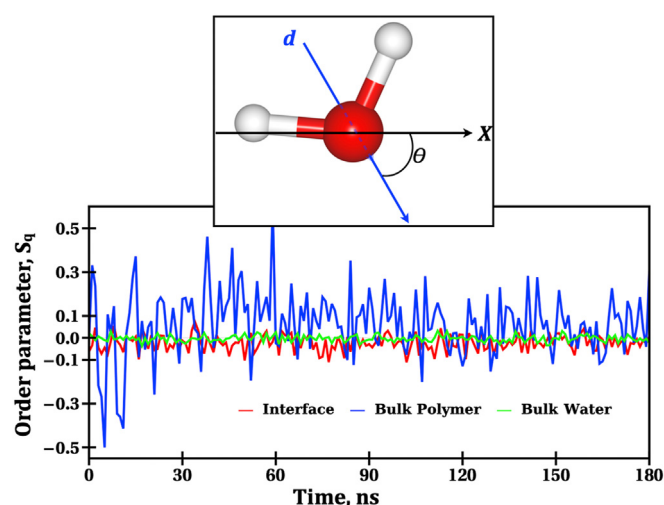


Fig. 5. Top: Schematic of the definition of order parameter with respect to x-axis and dipole d of water molecule. The drawn lines are added only as guide. Bottom: Orientational order parameter of water in different regions of MEP (221)/water system.

significant role in facilitating water diffusion in the epoxy network

[54]. Water molecules in the epoxy network can make different types of bonds. A random water molecule in the polymer network can form a hydrogen bond with another water molecule form cluster or with the epoxy network [21].

To elucidate the mechanism of diffusion of water molecules (cf. Fig. 3) and the advancement of entrapped water molecules in the epoxy network, we computed the number of H-bonds formed between water and different functional groups of epoxy networks. We first computed the total number of H-bonds formed between MEPs by the hydroxyl, amine, and ether functional groups. After that, the H-bonds between water-MEP hydroxyl, water-ether, and water-amine functional groups were accounted for. H-bonds formed by the different functional groups of MEPs were added to get the total number of H-bonds between water and MEPs. The details of the results are compiled in Fig. 7B. We have extracted the number of H-bonds for every 500 ps cycle from the 180 ns simulation trajectory. In all the cases, we observed that the significant H-bonds formed by hydroxyl groups of MEPs. We have computed average number of H-bonds formed by per donor by the different counterparts of the system and tabulated in Table 4.

A recent study reported that if the number of water molecules is higher inside or in the polymer vicinity, they formed aggregated clusters that are not big enough to show bulk water-like properties [19]. Those clusters are positioned in the micro void of the epoxy

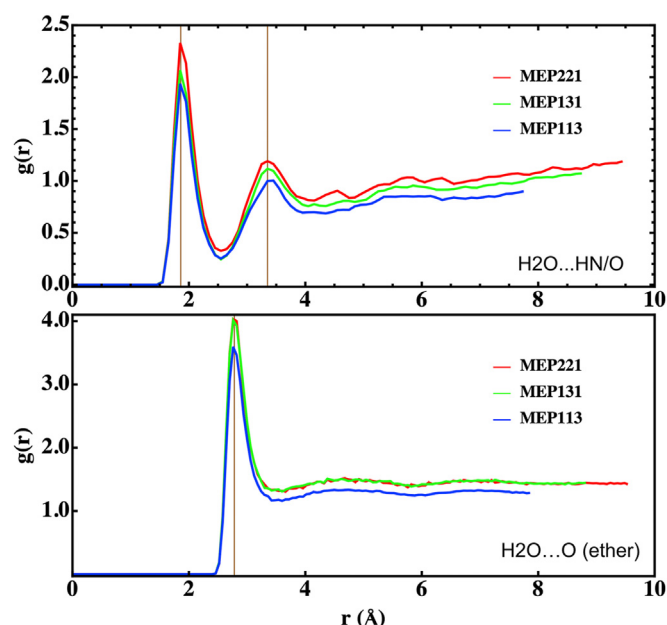


Fig. 6. Computed Radial distribution function (RDF) MEP/water system. *Top:* RDF of O...H pairs (oxygen of H₂O molecules and hydrogens of N and O) and *Bottom:* RDF of O...O pairs (Oxygen of ether and H₂O molecules). Brown lines point out the peak positions of the RDF. (For interpretation of the references to colour in this figure legend, the reader is referred to the Web version of this article.)

network and could influence the intramolecular interaction of MEPs network, which helps to swell the polymers. MEP network is composed of the hydrophobic component in the backbone like carbon atoms or alkyl side groups, which may impede the diffusion

Table 4

Average number of H-bonds per donor of the water/epoxy system.

	MEP (221)	MEP (131)	MEP (113)
MEP-MEP	10.41	10.30	9.91
MEP-Water	2.23	2.10	1.96
Water-Water	92.12	92.05	92.54

of the water molecules. From Fig. 7A, it can be noticed that the initial number of water molecules in the bulk polymer is lower and the concentration of water molecules increased as water molecules diffuse inside the polymer matrix. This phenomenon could be due to the exposure of the hydrophilic part of the MEP network to water molecules. Also, the MEP network creates various sizes of voids ranging from atomistic scale to the macroscale and is close to the interface, which facilitated the water accumulation [18]. This aggregation of random water molecules leads to the formation of clusters/strips, which could be broken down after a specific time followed by diffusion to the bulk polymer. The diffused water is stabilized by the hydroxyl groups that form H-bonding, as we discussed in the previous section. Three steps facilitate this process: (1) different orientation of water molecules at the interface due to the interaction with polymer molecules, (2) water stripes formed in the interface (3) collapse of water stripes and diffusion of water molecules into the polymer matrix.

The number of water molecules present at the interface and the bulk polymer matrix governs the structure of water in the respective regions. Quantity of water molecules determines the clustering of water, hydrogen bond formation, and the bond dynamics. Fig. 7 outlined the estimated number of water molecules at different regions of the system (Fig. 7A) and the number of hydrogen bonds between water and MEP (Fig. 7B). Diffusion of water molecules into polymer matrix overcomes the

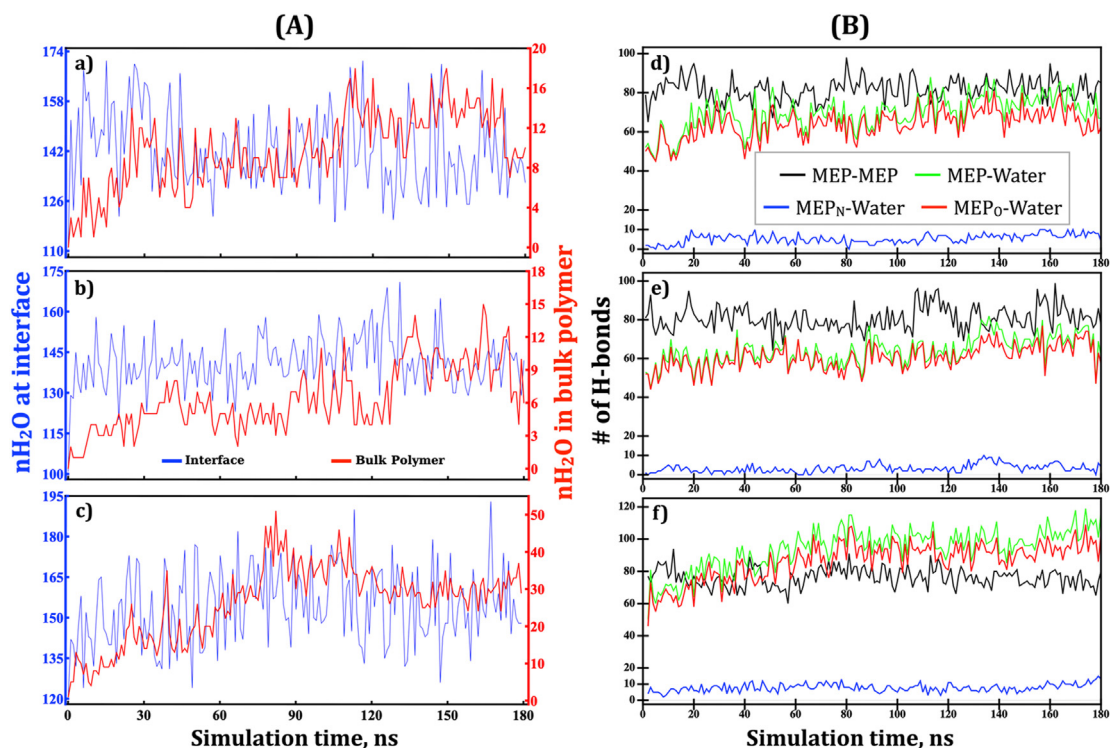


Fig. 7. Computed number of water molecules at polymer/water interface and in bulk polymer matrix, considering GDS and roughness of the polymer (A) and the number of hydrogen bonds formed by different counter-part of the polymer/water system as a function of time (B). MEP (221) [a] & d), MEP (131) [b] & e)] and MEP (113) [c] & f)]. The data points were collected for every 500 ps from the 180 ns trajectory.

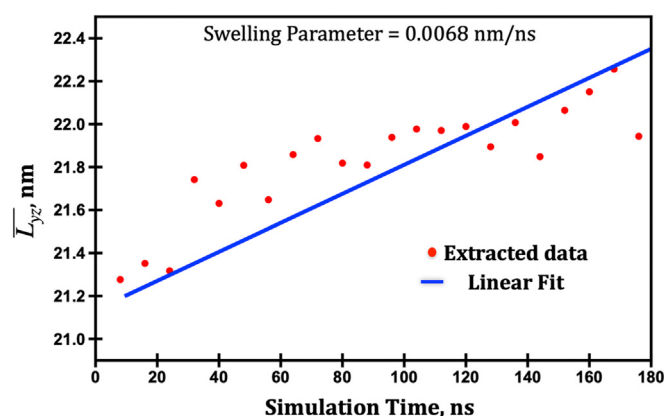


Fig. 8. Change of average lattice parameter in y and z direction ($\overline{L}_{yz}$) due to diffusion of water from bulk-to-interface and then to polymer matrix. The linear fit parameter or “swelling parameter” is 0.0068 nm/ns of MEP (221).

hydrophobicity that arises from the alkyl chain at the periphery of the interface as well the roughness of polymer surface. It is clear from Fig. 7A that the number of water molecules is larger in the case of MEP (113), followed by the MEP (221) and MEP (131), which might affect the surface roughness of MEPs' (Table 1). In the epoxy polymer matrix, diffused water molecules stabilized with the formation of hydrogen bonds by three different potential sites—hydroxyl, ether, or nitrogen groups. To elucidate the packing nature of the water molecules around the epoxy functional groups, we computed the radial distribution function (RDF) of water molecules in the vicinity of H-bonding sites of the epoxy polymer network.

3.5. Swelling behavior

We extracted the polymer lattice details throughout the simulation time to explore how the diffusion of water molecules affects the lattice parameters of the simulation cell. The polymer lattice changes along the yz plane are plotted in Fig. 8 (cf. Supporting Information, Figure S7) as a function of simulation time. We actually computed the average of the b and c lattice vectors and called it $\overline{L}_{yz}$ (although we use isotropic pressure parameters, the lattice parameters along y and z deviate in a small amount, and therefore we decided to consider the average). The fitted swelling parameters of MEP (221), MEP (131), and MEP (113) are 0.0068 nm/ns, 0.0058 nm/ns, and 0.0022 nm/ns, respectively. Although MEP (221) and MEP (113) are structurally symmetric however MEP (221) is denser compared to the MEP (113). It could be the one of the reasons of a higher swelling parameter revealed for MEP (221). Moreover, In the case of MEP (221), the roughness observed after 30 ns fluctuates to within less than 0.15 nm (cf. Figure S5). If one compares these fluctuations with the change of average lattice parameter in the y and z direction ($\overline{L}_{yz}$), the ratio is one order of magnitude smaller (cf. Fig. 8 and Figure S5.), and we conclude that the overall trend of the swelling parameter has small influence of the roughness fluctuations. This conclusion is also justified for the MEP (131) and MEP (113) cases, since the observed roughness after 40 ns is less than 0.1 nm (cf. Figure S5 and Figure S7).

4. Conclusion

We have carried out molecular dynamics simulation of Medipacs Epoxy Polymers (MEPs)/water interface. We build the possible polymer models, based on the experimentally suggested chemical formula and the interfacial properties MEPs/water explored.

Mathematical plane, Gibbs dividing surface (GDS), were used to model the interface. We have performed 180 ns all atom MD simulation and different physical properties were computed at the interface, in bulk water, and in bulk epoxy matrix. Order parameter, surface roughness, RDF, hydrogen bonds, diffusion of water and swelling behavior were computed for the MEP/water interface. Partial mass density profile of the MEP/water interface confirmed the diffusion of water inside the epoxy matrix. Water order parameter characterized the different orientation of water molecules at interface and in epoxy network. H-bonds formation in the MEPs networks with hydroxyl firms the stability of those water molecules in the polymer matrix. At the same time, H-bonds between MEP chains disrupt, leading to formation of H-bonds by the entrapped water molecules. This phenomenon allows the polymer structure to undergo a swelling process. We also estimated the swelling parameter from the average change in lattice parameter of the simulation box. The simulation results suggest that the roughness of the polymer 2D surface governs the number of water molecules at the water/polymer interface and in the bulk polymer. Also, the structural symmetry and packing properties of the polymer model regulates the swelling properties. The outcomes described here could relate the microscopic information obtained at the nanoscale to the material behavior at the macroscale. The understanding from this study could use to predict the global performance of the epoxy-based products at multiple length scales.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Supplementary data

Supplementary data to this article can be found online at <https://doi.org/10.1016/j.jmngm.2021.107915>.

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